TakeSure

Confidence before taking something

An Ayurveda × Allopathy herb–drug interaction checker for Indian patients, doctors and the Ministry of Ayush

Health & Well-Being · AI4Impact · SDG 3 · 9 · 10 · Web / PWA + API · Hindi · English

HACK@BVP 8.0 · Abstract Round
Team Bug Hunters
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1 · PROBLEM

- **Two medical systems, zero cross-checking.** A thyroid patient on Thyronorm adds ashwagandha; a transplant patient on tacrolimus drinks giloy kadha; a cardiac patient on warfarin takes guggul.
- **Each pair can cause harm:** ashwagandha can raise thyroid hormone, giloy can oppose immunosuppression, guggul can increase bleeding.
- **Nobody asks, nobody tells.** The doctor rarely asks about herbs, the patient rarely mentions them, the chemist has no tool, the label says nothing.
- **Existing tools do not cover it.** Medscape, Drugs.com and 1mg check only allopathic × allopathic pairs; Western herb databases are English-only, paid and do not know Indian brands like Himalaya Ashvagandha or Dabur Giloy Ghanvati.

37%	65%	19%	1,473
of hospital patients use allopathic and herbal medicines together	of herbal users do not know the dose, side effects or interactions	of Indian patients ever disclose herbal use to their doctor	ASU&H adverse-event reports received nationally since 2018

2 · PROPOSED SOLUTION

- **Scan both.** The patient photographs the allopathic prescription and the Ayurvedic product label (or types the names); every name is confirmed before checking.
- **Match Indian names.** Brand → generic (Thyronorm → levothyroxine) and product → herbs (Chyawanprash → amla, ashwagandha…) with Sanskrit / Hindi / Latin synonyms.
- **Decide from data.** A rule engine checks every herb–drug pair against a curated, cited knowledge base: GREEN / AMBER / RED / NOT IN DATABASE, with timing advice.
- **Explain simply.** AI rewrites only the matched evidence into plain Hindi or English, with sources and three questions to ask the doctor.
- **Report in one tap.** A one-page PDF for the doctor and a pre-filled suspected-ADR report in the Ministry of Ayush format for the Ayush Suraksha portal.

Our rule: The verdict comes from published data; AI only explains it.

AMBER — **Ashwagandha + Levothyroxine**
Use with caution · evidence grade B

Product: **Himalaya Ashvagandha** Rx: **Thyronorm 50**

Ashwagandha may raise thyroid hormone levels; the levothyroxine dose may need review. Do not stop either medicine on your own.

1. Should my TSH be rechecked in 6–8 weeks?
2. Is there a safer option for stress?
3. Should I keep a gap between the two?

Source: Sharma AK et al., J Altern Complement Med 2018 · PMID 28829155 · KB row v1.3 · verdict unchanged with AI off

Share PDF with doctor | Report suspected ADR

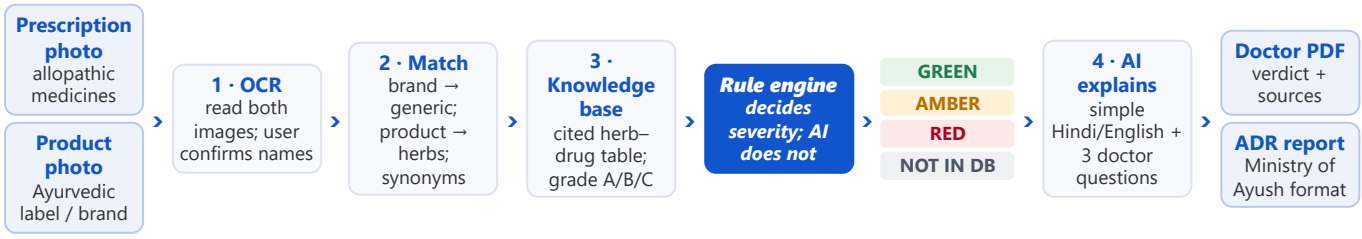
Concept of the result card — what the patient would see

WHY IT IS DIFFERENT

Feature	Medscape · Drugs.com · 1mg	MSKCC · Natural Medicines	TakeSure
Ayurveda × allopathy pair check	✗	~ single herbs	✓ every pair
Indian brands & Hindi	~	✗	✓ brands + synonyms
Source on every verdict	✓	✓	✓ + evidence grade
ADR report to Ministry of Ayush	✗	✗	✓ one tap

Our assessment from each tool's public documentation.

3 · APPROACH — WORKFLOW



Verdict rules: Worst row wins; an unknown pair is never shown as green; grade-C evidence caps at AMBER; the answer is identical with the AI switched off.

STACK & KNOWLEDGE BASE

Next.js + TypeScript PWA (Hindi/English UI) · FastAPI + Pydantic REST API · PostgreSQL in production, SQLite offline · Tesseract OCR · RapidFuzz matching (≥85 accept, 70–85 ask the user) · Groq LLM, JSON-schema output, temperature 0 · ReportLab PDFs. No model training; runs fully offline on a laptop.

40–60 high-frequency herb × drug pairs (15 common herbs × 12 drug classes) curated from PubMed, NIH LiverTox, MSKCC About Herbs, WHO monographs and the Ayurvedic Pharmacopoeia of India, graded on the Stockley's model. Decision support only: "ask your doctor", never diagnosis or dosing.

4 · EXPECTED IMPACT

- **Patients & caregivers.** A sourced answer before the first dose, in their own language, instead of silence or a guess. Free for citizens.
- **Doctors, vaidyas & chemists.** Finally learn what patients actually take; a QR on the chemist's bill or an API inside hospital and telemedicine systems makes the check routine.
- **Ministry of Ayush.** Structured, reproducible ADR reports and real co-use data from the catchment of its 99 peripheral pharmacovigilance centres.
- **Novelty.** Nothing public today checks an Ayurvedic product against an allopathic prescription. TakeSure builds the missing bridge table: Indian brands × Ayurvedic herb synonyms × graded evidence, open and auditable. Honest by design: grades, timing gaps instead of alarms, "not in database" instead of guesses.